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Joint optimization of empty container repositioning and inventory control applying dynamic programming and simulated annealing

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HIGHLIGHTS

• A joint optimization model is constucted based on the public and exclusive hinterlands within the regional port cluster.
• Different inventory control strategies are tested under (non)stationary demand.
• Periodic strategies better reducek total costs in nonstationary conditions.

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ABSTRACT

Based on the division of public and exclusive hinterlands, this paper studies the joint optimization problem of empty container repositioning and inventory control under non-stationary demand within the port cluster. This paper established a stochastic mixed integer programming model using distributed robust optimization methods, combined with quantitative and periodic inventory control strategies. After conducting a deterministic transformation, this paper designed a hybrid algorithm of dynamic programming and simulated annealing to solve the model, and compared the various costs under stationary and non-stationary scenarios. The results show that the joint optimization of empty container inventory control and repositioning can always reduce the total cost of empty container management for shipping companies. Periodic inventory control strategies are more suitable for shipping companies to apply to empty container management in non-stationary situations. Sensitivity analysis shows that there is a positive correlation between uncertain demand parameters and the total cost of shipping companies. The superiority of the empty container repositioning mode proposed in this paper under sea-land coordination has also been demonstrated by changing the accessibility parameters between terminals.

1. Introduction

Economies of scale produced by port integration have been demonstrated globally as port managers and governments have gradually increased the degree of port integration in multiport regions to promote economic efficiency [1]. China's ports are currently in a regional integration stage, promoting the coordinated regional development and transformation and upgrading of ports through key initiatives. In port integration, ports and inland container terminals jointly constitute an empty container repositioning network in the context of port clusters. Owing to fuzzy and uncertain boundaries, port hinterlands often cross-overlap; thus, the public hinterland prevails. Within the scope of the public hinterland, the radiation range and influence of ports have overlapping and competing characteristics, and container terminals can serve, and provide empty containers for, multiple ports simultaneously.

Shipping companies have always faced the problem of empty container repositioning, earning no profits from this process. In fiercely competitive shipping markets, scientific decision-making regarding empty container repositioning volume to reduce management costs is important for shipping companies to increase profits. Containers are both carrier and transportation goods: the full container state reflects the carrier attributes, whereas the empty container state reflects the transportation attributes. Therefore, repositioning empty containers involves both transportation and storage. Full containers are the supply of empty containers, and extant theories pay little attention to optimizing the empty container supply structure [2]. In the transportation link, empty

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and full containers require nearly identical transportation costs and time; however, unlike the transportation of full containers, that of empty containers generates no revenue. In the storage link, empty containers also occupy the same amount of space and requires basically the same storage costs. Essentially, the joint optimization of empty container repositioning and storage faces an uncertain environment in practical operations, originating from the uncertainty of empty container demand. In uncertain environments, empty container demand can be divided into stationary and non-stationary states. During a considerable period of decision-making, the empty container demand in certain decision-making cycles will follow certain laws and remain stable. However, in some periods, the empty container demand often remains non-stationary, making the specific distribution patterns difficult to grasp. Therefore, shipping companies must master scientific repositioning decisions related to the non-stationary demand for empty containers.

The remainder of this paper is organized as follows. Section 2 reviews the literature related to joint optimization of repositioning and inventory control. Section 3 describes our optimization problem in two phases. Then, we describe the assumptions and notations and develop a stochastic decision model for joint optimization in Section 4. Although this cannot be solved directly, we transform it into deterministic model in Section 5. After transformation, the new model can be solved using a hybrid algorithm, which we present in detail in Section 6. In Section 7, numerical experiments are conducted to evaluate the model and algorithm effectiveness. Section 8 summarizes the paper and provides recommendations for future research.

2. Literature review

2.1. Joint optimization of empty container repositioning and storage

Several achievements have been made in joint optimization research.

From the perspective of the type of joint optimization model constructed, Yang et al. [3] developed a two-stage game model to investigate the effects of the empty container movement problem and information asymmetry on system performance for optimal decision-making. Kamal [4] constructed a causal mechanism model from the shipping company perspective, and analyzed the impact of factors such as trade imbalance and seasonal demand. Xu et al. [5] designed a reinforcement learning framework with a multi-objective reward function to enhance container utilization and reduce total costs. Wang et al. [6] considered the container supply and shortage port as a whole and based on queuing theory an empty container repositioning was established minimizing the total cost of the empty container inventory and repositioning. Zhou et al. [7] developed a two-stage stochastic planning model of empty container transshipment and a separable segmented linear learning algorithm to continuously approximate the expectation function. Rajeswari et al. [8] used fuzzy inventory model to attain optimal length of the screening, and the repositioning cycle and the leasing cycle. Abdelshafie et al. [9] proposed a maritime empty container reposition modelling framework by integrating the agent-based modelling paradigm to help minimize the costs of empty container movement.

From the joint optimization perspective, Najafi and Zolfagharinia [10] used price management to control empty container repositioning and maximized company profits. Lee and Moon[11] proposed a robust formulation for foldable empty container repositioning considering non-stationary demand. Chi et al. [12] studied the volume of empty containers repositioned from surplus area to the shortage area and inventory of each node using sea, road and railway as transportation modes. Siswanto et al. [13] investigated multiple time windows and designed heuristic algorithms to co-optimize a ship's path selection and the number of containers to be loaded and unloaded in shorter computation time. Castrellon et al. [14] assessed two ECR strategies via dry ports-street turns and extended free temporary storage with different levels of container substitution.

Currently, models constructed for joint optimization are mainly two-stage or game models, considering factors such as price management, revenue maximization, service level, cooperation and sharing, and route selection during optimization. Regarding scope, most research focuses on the portside and sea-facing hinterlands, with constructed transportation networks mainly revolving around maritime transportation. The current research perspective has clearly not progressed beyond the limitations of individual seaports, and a larger range of port clusters has not received attention. Further, the landward hinterland has not been sufficiently linked to ports. Therefore, in model construction of current studies, the empty container supply from the landward hinterland of the port is ignored, and the direction of empty container repositioning only exists between ports or terminals (i.e., street-turn mode).

2.2. Joint optimization for non-stationary demand

In actual empty container repositioning, the supply and demand of empty containers are in a non-stationary state, and the supply and demand laws of empty containers are difficult to accurately portray. Therefore, research on empty container repositioning is often based on the non-stationary situation of the demand for empty containers.

From the perspective of handling non-stationary demand, Rodrigues et al.[15] examined sailing time uncertainty and considering methods for centralized handling of uncertainty (i.e., deterministic models with inventory buffers, robust optimization, and stochastic planning). Liu et al. [16] proposed a two-stage distribution robust optimization method, combining it with Pareto optimality for large-scale arithmetic cases. Xiang et al. [17] proposed that an arbitrary supply of empty containers depended on laden containers shipped and turnover times. By accounting for turnover times, periodically repositioning empty containers was studied. Misra et al. [18] proposed hybrid time discretization, combines continuous time with a multigrid discretized timeframe, and proposed a rolling time-domain strategy to solve the problem. Cai et al. [19] designed a data-driven framework for empty container repositioning based on existing large datasets, and used machine-learning methods to predict the supply and demand of empty containers. Feng et al. [20] studied that trip-sharing mode can reduce the transportation costs of repositioning under certain demand. Liang et al. [21] developed a stochastic dynamic programming model to jointly optimize the slot allocation and empty container repositioning based on current container stocks, realized demand and the mean demand of the future stages.

Considering inventory processing methods under non-stationary demand, Song et al. [22] designed a two-stage particle swarm optimization algorithm, and provided ports with a method for determining the upper limit of empty container inventory. Soroush et al. [23] adopted three demand distributions and used chance constrained programming to transform them into an accurate mixed-integer nonlinear programming. Song and Dong [24] examined the repositioning process under uncertainty of the destination port and proposed a method to set a retention threshold for empty containers at port. Shaabani et al. [25] considered that a ship's route is deterministic and inventory at the port must remain within a given range, and designed heuristic algorithms for solution. Sarmadi et al. [26] explored dynamic decision-making for container transportation and inventory in uncertain environments.

Joint optimization under non-stationary demand primarily involves optimization methods such as stochastic programming, robust optimization, machine learning, and value iteration. Regarding empty container inventory, some studies directly minimize storage costs but do not use inventory control strategies. Some studies have involved buffer inventory, threshold control strategies, and periodic steady strategies. Current inventory strategies are all pre-made and relatively single. Inventory of empty and full containers is constantly changing as they are transported, and the role of inventory control strategies varies across practical application scenarios. However, previous studies have not found a close relationship between empty container inventory and actual numbers of empty and full containers transported each week Thus, the applicability of various inventory control strategies in different situations remains unverified.

2.3. Research gap

(1) The existing literature focuses more on the coastal hinterland and is still limited to the direct supply of hinterlands to ports in terms of geographical location. The integration of Chinese ports has led to many overlapping phenomena between port clusters and inland areas, resulting in the emergence of public and exclusive hinterlands. However, there has not been sufficient research on the interaction between empty containers in the new type of hinterlands and ports. The repositioning and storage of empty containers within the public hinterland, exclusive hinterland, and between ports within the port cluster are issues that require special attention. Therefore, the starting point of this study is the public and exclusive hinterlands within the port cluster, and the optimization decision of empty container repositioning volume and inventory level is made through the coordinated transportation mode of land and sea.

(2) In the research on non-stationary demand for empty containers, existing literature mostly focuses on selecting inventory control strategies to reduce the total cost of empty container repositioning. However, for shipping companies, they sometimes face non-stationary demands and sometimes face stable demands during the decision-making cycle. Different inventory control strategies will have different control effects under different demand situations, and shipping companies should flexibly choose empty container inventory control strategies based on actual situations. Therefore, it is necessary to conduct in-depth exploration on the applicability of different inventory control strategies in different demand situations. This paper investigates the applicability of quantitative inventory control strategies and periodic inventory control strategies under different demand situations, in order to provide decision support for the selection of inventory control strategies for shipping companies.

2.4. Major contributions

According to the inductive summary of the current paper, this study's main contributions to the joint optimization of empty container repositioning and inventory are described below.

(1) In the context of port integration, we divide the hinterlands of the ports into public hinterland and exclusive hinterland. Container terminals within public hinterlands can provide empty containers for multiple ports within their coverage area, while terminals within exclusive hinterlands can only provide empty container supply for one port. Therefore, we proposed the "single terminal to multiple ports" empty container supply mode, and also constructed a sea-land collaborative empty container transportation network within the port cluster, which enables full interaction between ports landward hinterlands and port clusters, providing shipping companies with a new method for supplying empty container resources.

(2) We introduce two inventory control strategies, namely quantitative inventory control strategy and periodic inventory control strategy, and verify the applicability of the two inventory control strategies under different demand situations through the construction of a joint optimization model for repositioning and inventory control. Simultaneously, the inventory threshold calculation formula designed in this study fully associates the empty container inventory at each period with the actual empty container inflow and outflow. By analyzing the cost values of different inventory control strategies in various situations, the applicability of both inventory control strategies is verified, providing decision-making support for shipping companies when selecting inventory strategies in non-stationary demand situations.

3. Problem formulation

In the hinterland of the port cluster, public and exclusive hinterlands are divided. Public hinterlands refer to the hinterland shared by two or more ports, that is, the part where the attraction range of several ports overlaps with each other. With the rapid development of inland railway and highway construction in China, the public hinterlands are gradually expanding. Generally speaking, the public hinterland is a crossing area with competition characteristic. Full containers at inland container terminals within a public hinterland can choose to sail from the multiple ports they cover. Exclusive hinterlands are independently owned by a port without competition. The exclusive hinterland is also the direct hinterland of the port, which is the hinterland of the port itself. All goods required for sea transportation in this area must pass through this port. Empty container repositioning can only be performed between inland container terminals within the same port coverage area. In port clustering, empty container repositioning can occur between various ports within a cluster, terminals within the same port coverage area, or ports and either inland or exclusive public terminals.

Fig. 1 shows four ports, A, B, C, and D, within the port cluster, with eleven inland container terminals. Terminals 3, 6, 7, 9, and 10 are exclusive inland terminals for each port; the others are public inland terminals between two or more ports. Taking Ports A and D as examples, the inland terminals covered by Port A include 1, 2, 6, and 11, while those covered by Port D are 8, 9, 10, and 11. A port and its respective covered terminals carry out cyclic empty and full container transportation. Simultaneously, empty and full containers flow between inland terminals under the same port coverage. When the demand for empty containers at port or inland terminal nodes cannot be met, they can be transported from terminals under the same coverage or directly from the port, such as between Terminals 6 and 11 or Terminals 11 and 10. However, inland terminals located within different port coverage areas, such as those between Terminals 6 and 10, cannot conduct empty container repositioning. In addition, full containers located in the public hinterland of Terminal 11 can be transported from Ports A and D out to sea. However, Terminal 6 is the exclusive hinterland of Port A; therefore, full containers can only be shipped from Port A, and Terminal 9 is the same as Terminal 10.

In an empty container repositioning network such as that described above, for each port node within the cluster, the empty container inflow may come from the port's remaining inventory from the previous period, repositioning volume of other ports whose transportation for the previous period arrived in the current week, repositioning volume of the public hinterland terminal, repositioning volume of the exclusive hinterland terminal, conversion volume of full containers in the previous period, or renting volume, while the empty container outflow volume includes the remaining empty containers volume of the current period, and the volume of the empty containers repositioned to other ports. The outflow of empty containers includes the remaining empty containers in the current period; empty containers repositioned to other ports, public inland terminals, or exclusive inland terminals; and full containers shipped out of the port.

For the terminal nodes, the empty container inflow includes the remaining empty container inventory of the terminal in the last period, conversion of full containers delivered by the port in the last period, repositioning of empty containers from the port in the last period or between terminals under the coverage of the same port, conversion of full containers, and amount of renting containers. The empty container outflow includes the empty container inventory in the current period, empty containers repositioned to the port and delivered to the port or terminals under the coverage of the same port, empty containers delivered to the port, and renting containers. The empty container outflow includes the empty container inventory in the current period, volume of empty containers repositioned to the port and other terminals covered by the same port, and volume of full containers shipped to the port and other terminals (Fig. 2).

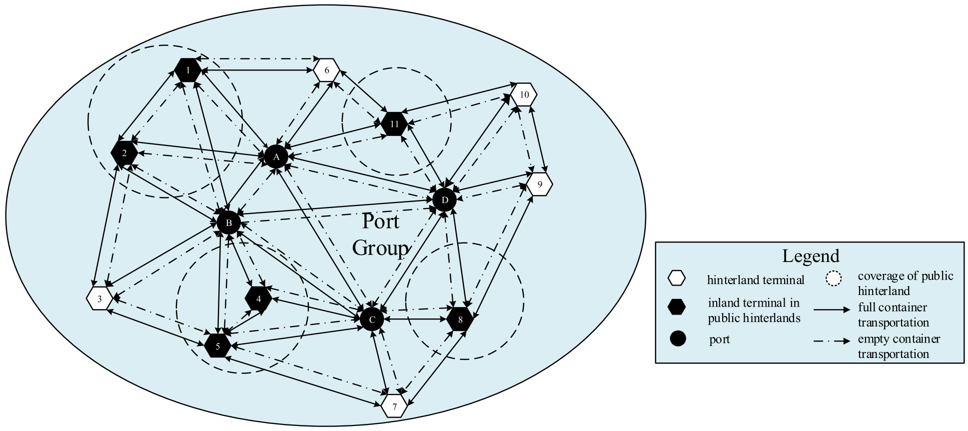

Fig. 1. Schematic diagram of empty container repositioning network within the regional port cluster.

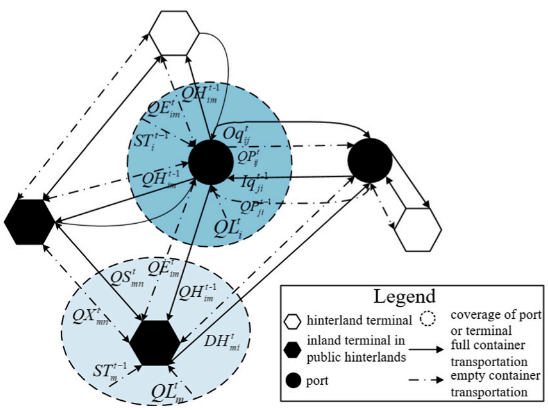

Fig. 2. Schematic illustration of the inflow and outflow of empty containers at port and terminal nodes within a regional port cluster.

In this study, with the objective of minimizing the total cost of all periods, the constraints such as the balance of the inflow and outflow of empty containers at each port node, balance of the repositioning volume between ports and public hinterlands, terminals of exclusive hinterlands, balance of the repositioning volume between terminals in public hinterlands, limit of the stockpile capacity, and limit of the transportation capacity are comprehensively considered. Simultaneously, an interval of holding inventory is introduced to rationally control the empty containers of each port in the cluster. The distributed robust optimization method is adopted to handle the random constraints. The random chance constraints are transformed into a deterministic problem by introducing a set of probability distributions for solving the problem by using the combination of dynamic programming and simulated annealing algorithms as well as heuristic rules. The aims are to demonstrate the superiority of joint optimization, and analyze the effectiveness of quantitative and periodic inventory control strategies in controlling total costs under non-stationary demand conditions.

4. Mathematical model

4.1. Assumptions

(1) Containers have dual attributes. In the full container state, they reflect the transport vehicle attribute, while in the empty container state, they reflect the general transport cargo attribute. After the cargo is unloaded, the full container becomes an empty container and can be used as the supply of empty containers for the current period. Usually, liners operates on a weekly basis, while full and empty container transportation typically circulates at a frequency of one week. Based on the frequency of liners, this paper assumes that empty containers repositioned by sea between ports in the previous period arrive at the port in this period as the supply of empty containers in this period. The empty containers repositioned between the port and terminals in the previous period will arrive in this period and be supplied as empty containers. Empty containers repositioned between terminals will arrive in this period and be directly supplied as empty containers.

(2) No shipping companies can have containers stored at any of their demand locations, while container leasing companies can relatively meet their requirements and ensure the supply of containers as much as possible. The rental business has developed rapidly, with rental companies accounting for over 45 % of the world's total container volume. In China, especially in the container fleets of offshore and domestic shipping companies, leased containers account for over 90 % of the total container volume. Therefore, this paper assumes that renting containers can arrive within the current period, without considering the terms related to returning the containers.

(3) The focus of this paper is on the joint optimization decision of empty container repositioning and inventory within the port cluster. The transportation mode is pre-set to combine sea and land transportation, and the selection of transportation mode is not considered in this paper. Based on the transportation radius of road transportation and railways (The economic radius of road transportation is usually within 200 kilometers), railway transportation is assumed to be used if the distance between the port and public hinterland exceeds 200 km; otherwise, road transportation will be used.

(4) Only Twenty-foot Equivalent Unit (TEU) containers are considered, without considering other special types of containers.(Table 1 and Table 2)

4.2. Notation description

Table 3 shows the parameters used in this paper and their description.

4.3. Calculation of empty container inventory threshold

When the empty containers at port are in shortage, it is insufficient to meet the demand for empty containers, which will not only cause the loss of freight but also reduce the shipping company's service level. However, when the empty containers at port are a surplus, many empty containers will remain at port after the demand for empty containers has been met, which will generate extra storage cost for the shipping company. Therefore, setting a reasonable threshold value for empty containers is especially important for shipping companies at ports. The threshold should not only ensure that the demand for empty containers is met without many renting containers but also avoid many empty

Table 1
A classification of relevant papers in the literature.

Paper	Research scope	Model	Inventory control strategy	Solution method
[3]	Two-port system	Screening model	/	Two-part tariff contract
[4]	Maritime transportation	A novel model of the casual mechanism of ECR	/	Fuzzy bayes network method & gray box pooling solution
[5]	Maritime transportation	Multi-objective optimization	/	Markov decision process and reinforcement learning techniques
[6]	Maritime transportation	Joint optimization model	Birth-death inventory	Genetic algorithm
[7]	Maritime transportation	Two-stage stochastic programming model	/	Separable piecewise linear learning algorithm
[8]	Maritime transportation	Fuzzy inventory model	Inventory model	Simulation
[9]	Maritime transportation	Agent-based maritime logistics empty container redistribution model	/	Simulated annealing
[10]	Maritime transportation	Mathematical formulation	/	Robust pricing and quality setting strategy
[11]	Maritime industry	Robust formulation	/	Robust optimization
[12]	Multimodal transportation	Multi-period linear programming model	/	CPLEX12.10
[13]	Maritime transportation	Inventory routing optimization model	/	Multi-heuristics & genetic algorithm
[14]	Sea-land transportation	Multiparadigm simulation combined agent-based and discrete-event model	/	Simulation

Table 2
A classification of relevant papers in the literature.

Paper	Research scope	Model	Inventory control strategy	Solution method
[15]	Maritime transportation	Stochastic programming	Inventory buffs	Robust optimization
[16]	Maritime transportation	Two-stage distributional robust optimization	/	Benders decomposition
[17]	Maritime transportation	Joint optimization	/	Novel branch-and-price algorithm
[18]	Maritime transportation	Novel hybrid time representation	/	Heuristic algorithm
[19]	Rail transportation	Data-driven framework for the ECR	/	Machine learning
[20]	Maritime transportation of bulk cargo	Three business modes and supply chain coordination	/	Simulation
[21]	Maritime transportation	Stochastic dynamic programming	/	Approximate dynamic programming
[22]	Maritime transportation	Empty container repositioning and non-repositioning model	/	Two-stage particle swarm optimization algorithm
[23]	Maritime transportation	Stochastic optimization model	Pre-specified lower and upper storage levels	Chance-constrained programming
[24]	Container shipping	Stochastic optimization model	Threshold control policy	Simulation
[25]	Maritime transportation	Mixed integer programming	/	CPLEX
[26]	Dry port network	Two-stage stochastic programming model	Location-allocation of dry ports	Robust sample average approximation & benders decomposition
this paper	Port cluster	Mixed stochastic integer programming	Periodic and quantitative inventory strategies	Distributed robust optimization & dynamic programming &SA

containers being amassed at port. Among the inventory control strategies, the quantitative inventory control strategy can fulfil these requirements.

Set the following new notations:

μ_{ij}^t : Mean value of empty containers shipped from port $i \in P$to port $j \in P$in period $t \in T$ (i.e., demand for empty containers);

μ_{ji}^t : Mean value of empty containers shipped from port $j \in P$to port $i \in P$in period $t \in T$ (i.e., supply of empty containers);

μ_{im}^t : Mean value of empty containers shipped from port $i \in P$to terminal $m \in H$in period $t \in T$;

σ_{ij}^t : Standard deviation of empty container demand at port $i \in P$in period $t \in T$, $\forall j \in P$;

D_i^t : Extremum of empty containers repositioned at port $i \in P$in period$t \in T$ (i.e., minimum inventory level);

U_i^t : Extremum of empty containers repositioned at port $i \in P$in period$t \in T$ (i.e., maximum inventory level).

When the quantity of empty containers stored at port is not within the threshold interval, the empty container inventory needs to be made to reach the range of the interval through repositioning and renting containers, or repositioning containers at other ports in the cluster and public inland terminals within the coverage area, to reduce the quantity of empty containers at port. The calculations are shown in Eqs. (3) and (4):

$$U_i^t = \sum_{j\in P} \mu_{ij}^t + \sqrt{\sum_{j\in P} \left(\sigma_{ij}^t\right)^2} \tag{3}$$

$$D_i^t = \max\left\{0, \left(\sum_{j\in P}\mu_{ji}^t - \sum_{j\in P}\mu_{ij}^t - \sum_{m\in H}\mu_{im}^t\right)\right\} + \sqrt{\sum_{j\in P}\left(\sigma_{ij}^t\right)^2} \tag{4}$$

Eq. (3) represents the calculation method for the upper limit of empty container inventory, using the sum of the mean and standard deviation of the total demand for empty containers in each period at port i. Eq. (4) represents the formula for the lower limit of empty container inventory, which uses the sum of the mean and standard deviation of the total difference (net outflow) between the inflow and outflow of empty containers at port$i$ per period. This is to enable empty containers to flow from the remaining container port to the shortage port and reduce reverse empty container transportation.

Liner transportation has strong periodicity, and its fixed frequency and running time make it more suitable for applying the periodic inventory control strategy. The quantitative inventory control strategy

Table 3
Notation description.

Notations		Description
Sets	T	Length of plan period , $t \in T$.
	P	Set of all ports in the cluster , $i, j \in P$.
	H	Set of inland terminals, $m, n \in H$.
Cost parameter	CP_{ij}	Unit transportation cost of empty container between port $i \in P$and port$j \in P$.
	$CS1_m$	Unit inventory cost of empty container of hinterland terminal $m \in H$.
	$CS2_i$	Unit inventory cost of empty container of port $i \in P$.
	CE_{im}	Unit transportation cost of empty container between port$i \in P$and hinterland terminal$m \in H$.
	CL_i	Unit renting cost of port $i \in P$.
	CL_m	Unit renting cost of hinterland terminal$m \in H$.
	CX_{mn}	Unit transportation cost of empty container between hinterland terminal $m \in H$and $n \in H$.
Random variables	Iq_{ji}^t	Number of full containers arriving at port $i \in P$ from port $j \in P$ in period$t \in T$.
	Oq_{ij}^t	Number of full containers shipped from port$i \in P$ to port$j \in P$ in period $t \in T$ (i.e., the current demand for empty containers at the port$i \in P$).
	QH_{im}^t	Number of full containers shipped from port $i \in P$ to hinterland terminal $m \in H$ in period$t \in T$.
	DH_{mi}^t	Number of full containers that arrived at port $i \in P$ from hinterland terminal $m \in H$ in period$t \in T$.
	ω_m^t	Difference in inflow and outflow of empty containers of hinterland terminal $m \in H$in period$t \in T$: $\omega_m^t = \begin{cases} IV1_m - \sum_{i \in P} \lambda_{im} DH_{mi}^t, t = 1 \\ \sum_{i \in P} \left[\lambda_{im} (QH_{im}^{t-1} - DH_{mi}^t)\right], t \geq 2 \end{cases}, \forall t \in T, \forall m \in H$ (1)
	ω_i^t	Difference in inflow and outflow of empty containers of port $i \in P$in period$t \in T$: $\omega_i^t = \begin{cases} IV2_i - \sum_{j \in P} Oq_{ij}^t, t = 1 \\ \left(\sum_{j \in P} Iq_{ji}^{t-1} - \sum_{m \in H} \lambda_{im} QH_{im}^{t-1}\right) - \sum_{j \in P} Oq_{ij}^t, t \geq 2 \end{cases}, \forall t \in T, \forall i \in P$ (2)
	α_m^t	(0, 1) variables. The value is 1 when$\omega 1_m^t > 0$, and 0 otherwise. Thus, the value is 1 when the terminal repositions containers outward, and 0 when it repositions containers inward or rents them.
	β_i^t	(0, 1) variables. The value is 1 when$\omega 2_i^t > 0$, and 0 otherwise. Thus, the value is 1 when the port repositions containers outward, and 0 when it repositions containers inward or rents them.
Other parameter	$IV1_m$	Initial stock level of hinterland terminal $m \in H$.
	$IV2_i$	Initial stock level of port$i \in P$.
	λ_{im}	1 if the hinterland terminal $m \in H$is in the port$i \in P$'s coverage area, and 0 otherwise.
	γ_{mn}	1 when the hinterland terminal $m \in H$ is reachable from hinterland terminal$n \in H$, and 0 otherwise.
	$W_m^{\max}$	Maximum empty container stock at hinterland container terminal$m \in H$.
	$W_i^{\max}$	Maximum empty container stock at port$i \in P$.
	F_{ij}^t	Maximum container capacity between ports $i \in P$ and $j \in P$in period$t \in T$.
	Cap_m	Maximum loading and unloading capacity of terminal $m \in H$.
	Cap_i	Maximum loading and unloading capacity of port $i \in P$.
Decision variables	QP_{ij}^t	Number of empty containers repositioned from port $i \in P$to port $j \in P$in period $t \in T$.
	QL_m^t	Number of renting empty containers at hinterland container terminal $m \in H$ in period $t \in T$.
	QL_i^t	Number of renting empty containers at port $i \in P$ in period $t \in T$.
	QE_{im}^t	Number of empty containers repositioned from port $i \in P$ to hinterland container terminal $m \in H$ in period $t \in T$.
	QX_{mn}^t	Number of empty containers repositioned from hinterland container terminal $m \in H$to hinterland container terminal $n \in H$ in period$t \in T$.
Derived variables	ST_m^t	Empty container inventory at hinterland terminal $m \in H$in period $t \in T$.
	ST_i^t	Empty container inventory at port $i \in P$in period $t \in T$.

requires frequent counting of empty container inventory; however, the periodic inventory control strategy only requires replenishing empty container inventory at a fixed time. Under this strategy, the time for empty container replenishment is fixed, and the amount of empty container replenishment is constantly changing. The formula is shown in Eq. (5).

Set the following new notations:

L : Lead time for empty container repositioning (i.e., time required for shipping companies to send empty container repositioning requests to other ports and inland terminals in advance);

α : Service level coefficient;

Z : Time interval for empty container repositioning (i.e., fixed supply time for empty containers);

$\sigma_{ij}^{\prime t}$: Standard deviation of empty container demand during the lead time of empty container repositioning at port$i \in P$ in period $t \in T$.

$$S_i^t = (Z+L)\left(\sum_{j \in P} \mu_{ij}^t + \sum_{m \in H} \mu_{im}^t\right) + \alpha \sqrt{\left(\sum_{j \in P} \sigma_{ij}^t\right)^2 (Z+L) + \left(\sum_{j \in P} \sigma_{ij}^{\prime t}\right)^2 \left(\sum_{j \in P} \mu_{ij}^t + \sum_{m \in H} \mu_{im}^t\right)} \quad (5)$$

Eq. (5) represents the maximum inventory of empty containers at port within each decision period. The first half of the formula is the product of the decision period, lead time, and sum of empty container demand for empty containers. The second half of the formula adjusts the fluctuation range of the threshold through the service level coefficient and standard deviation. The average demand for empty containers from ports to inland terminals is introduced into the formula, to allow the calculated threshold to meet the sum of empty container demand for ports and inland terminals as much as possible.

4.4. Joint optimization model for non-stationary demand

The objective function of the model is to minimize shipping companies' total costs in all the decision periods, which consists of three parts: C_1 is the cost of repositioning, including repositioning empty containers between ports, between ports and inland terminals, and between public inland terminals; C_2 is the cost of renting containers at ports and inland container terminals; and C_3 is the cost of storage, including storing empty containers at ports and inland terminals.

$$C_1 = \sum_{t \in T}\left(\sum_{i \in P}\sum_{j \in P}\beta_i^t CP_{ij} QP_{ij}^t + \sum_{i \in P}\sum_{m \in H}\alpha_m^t \lambda_{im} QE_{im}^t CE_{im} + \sum_{m \in H}\sum_{n \in H}\gamma_{mn} QX_{mn}^t CX_{mn}\right) \quad (6)$$

$$C_2 = \sum_{t \in T}\sum_{i \in P}\beta_i^t CL_i QL_i^t + \sum_{t \in T}\sum_{m \in H}\alpha_m^t CL_m QL_m^t \quad (7)$$

$$C_3 = \sum_{t \in T} \left(\sum_{m \in H} \max(ST_m^t, 0) CS1_m + \sum_{i \in P} \max(ST_i^t, 0) CS2_i \right) \tag{8}$$

$$\min C = C_1 + C_2 + C_3 \tag{9}$$

Subject to :

(1) Iteration of empty container inventory levels. Eqs. (10) and (11) represent the iterative relationship of empty container inventory at ports and inland container terminals at end-of-period. The overall calculation principle is to subtract the outflow from the inflow of empty containers.

In Eq. (10), for all inland terminal nodes, the inflow of empty containers in period t, the first half of the equation, includes the remaining empty container inventory of the previous period, volume of full containers shipped from the port to the terminal in the previous period, volume of empty container repositioning from the port to the inland terminal in the previous period, possible empty container repositioned volume from other terminals and possible renting container volume. The outflow of empty containers in period$t$, the latter half of the equation, includes volume of repositioning between terminals in the same coverage area, possible empty container repositioning volume to ports, and demand for empty containers at the inland terminal in the current period.

In Eq. (11), for all port nodes, the inflow of empty containers in period$t$, the first half of the equation, includes the remaining empty container inventory of the previous period, volume of full containers shipped from other ports in the previous period, volume of empty containers repositioned form other ports in the previous period, volume of empty containers repositioned from other hinterland terminals and possible renting container volume. The outflow of empty containers in period$t$, the latter half of the equation, includes the repositioning volume to other ports and hinterland terminals in the same coverage area in the current period, and volume of full containers shipped to other ports and hinterland terminals.

$$\begin{aligned} ST_m^t = {} & ST_m^{t-1} + \sum_{i \in P} \lambda_{im} QH_{im}^{t-1} + \sum_{i \in P} \beta_i^t \lambda_{im} QE_{im}^{t-1} + \sum_{n \in H} \alpha_n^t \gamma_{mn} QX_{nm}^t + \alpha_m^t QL_m^t \\ & - \alpha_m^t \left(\sum_{n \in H} \gamma_{mn} QX_{mn}^t + \sum_{i \in P} \lambda_{im} QE_{im}^t \right) - \sum_{i \in P} DH_{mi}^t, \forall t \in T, \forall m \in H \end{aligned} \tag{10}$$

$$\begin{aligned} ST_i^t = {} & ST_i^{t-1} + \sum_{j \in P} Iq_{ji}^{t-1} + \sum_{j \in P} \beta_j^t QP_{ji}^{t-1} + \sum_{m \in H} \alpha_m^t \lambda_{im} QE_{mi}^t + \beta_i^t QL_i^t \\ & - \beta_i^t \left(\sum_{j \in P} QP_{ij}^t + \sum_{m \in H} \lambda_{im} QE_{im}^t \right) - \sum_{m \in H} \lambda_{im} QH^{t-1} - \sum_{j \in P} Oq_{ij}^t, \forall t \in T, \forall i \in P \end{aligned} \tag{11}$$

(2) Equilibrium constraints on empty container inflow and outflow. Eqs. (12)–(15) represent the balance of empty container movements between ports and inland terminals. Taking Eq. (10) as an example, if the value of α_m^t is 0, then that the terminal lacks containers at that time; thus, repositioning containers to the outside is not possible. Therefore, the corresponding decision variable QE_{im}^t should have a value of 0, and then the value of $\sum_{m \in H} \alpha_m^t \lambda_{im} QE_{mi}^{t-1}$ in Eq. (13) should be 0.

$$\beta_i^t \left(\sum_{j \in P} QP_{ij}^t + \sum_{m \in H} \lambda_{im} QE_{im}^t \right) \le ST_i^{t-1} + \sum_{j \in P} Iq_{ji}^{t-1} - \sum_{m \in H} \lambda_{im} QH^{t-1} - \sum_{j \in P} Oq_{ij}^t, \forall t \in T, \forall i \in P \tag{12}$$

$$\begin{aligned} & \sum_{j \in P} \beta_j^t QP_{ji}^{t-1} + \sum_{m \in H} \alpha_m^t \lambda_{im} QE_{mi}^{t-1} + \beta_i^t QL_i^t \ge \\ & - \left(ST_i^{t-1} + \sum_{j \in P} Iq_{ji}^{t-1} - \sum_{m \in H} \lambda_{im} QH^{t-1} - \sum_{j \in P} Oq_{ij}^t \right), \forall t \in T, \forall i \in P \end{aligned} \tag{13}$$

$$\alpha_m^t \left(\sum_{i \in P} \lambda_{im} QE_{mi}^t + \sum_{n \in H} \gamma_{mn} QX_{mn}^t \right) \le ST_m^{t-1} + \sum_{i \in P} \lambda_{im} QH_{im}^{t-1} - \sum_{i \in P} DH_{mi}^t, \forall t \in T, \forall m \in H \tag{14}$$

$$\begin{aligned} & \sum_{i \in P} \beta_i^t \lambda_{im} QE_{im}^{t-1} + \sum_{n \in H} \beta_n^t \gamma_{mn} QX_{nm}^t + \alpha_m^t QL_m^t \ge \\ & - \left(ST_m^{t-1} + \sum_{i \in P} \lambda_{im} QH_{im}^{t-1} - \sum_{i \in P} DH_{mi}^t \right), \forall t \in T, \forall m \in M \end{aligned} \tag{15}$$

(3) Maximum stacking capacity constraints:

$$ST_m^t \le W_m^{\max} \quad , \forall t \in T, \forall m \in H \tag{16}$$

$$ST_i^t \le W_i^{\max} \quad , \forall t \in T, \forall i \in P \tag{17}$$

(4) Maximum transportation capacity constraints:

$$QP_{ij}^t + Oq_{ij}^t + \sum_{m \in H} \lambda_{im} DH_{mi}^t \le F_{ij}^t \quad , \forall i, j \in P, \forall t \in T \tag{18}$$

(5) Loading and unloading capacity constraints:

$$\sum_{j \in P} \beta_i^t QP_{ij}^t + \sum_{m \in M} \beta_i^t \lambda_{im} QE_{im}^t \le Cap_i, \forall t \in T, \forall i \in P \tag{19}$$

$$\sum_{i \in P} \beta_i^t QP_{ij}^t + \sum_{m \in H} \alpha_m^t \lambda_{jm} QE_{mj}^t \le Cap_j, \forall t \in T, \forall j \in P \tag{20}$$

$$\sum_{i \in P} \alpha_m^t \lambda_{im} QE_{mi}^t + \sum_{n \in H} \alpha_m^t \gamma_{mn} QX_{mn}^t \le Cap_m, \forall t \in T, \forall m \in M \tag{21}$$

$$\sum_{i \in P} \beta_i^t \lambda_{in} QE_{in}^T + \sum_{m \in H} \alpha_m^t \gamma_{mn} QX_{mn}^t \le Cap_n, \forall t \in T, \forall n \in H \tag{22}$$

(6) Non-negative constraints on decision variables.

$$QP_{ij}^t, QL_i^t, QL_m^t, QX_{mn}^t, QE_{im}^t \ge 0, \text{asinteger} \tag{23}$$

Based on the above constraints, the following constraints are added to form a joint optimization model for empty container repositioning and storage under the consideration of quantitative inventory control strategy:

$$D_i^t \le ST_i^t \le U_i^t, \forall i \in P, \forall t \in T \tag{24}$$

$$\sum_{m \in H} \alpha_m^t \lambda_{im} QE_{im}^t + \sum_{j \in P} \beta_j^t QP_{ij}^t \le \max\{ST_i^t - U_i^t, 0\} \; , \forall t \in T, i \in P \tag{25}$$

$$\sum_{m \in H} \alpha_m^t \lambda_{im} QE_{mi}^{t-1} + \sum_{j \in P} \beta_j^t QP_{ji}^{t-1} + \beta_i^t QL_i^t \le \{D_i^t - ST_i^t\}, \forall t \in T, i \in P \tag{26}$$

Eq. (24) indicates that the port's current empty container inventory should be controlled within the threshold range. Eq. (25) indicates that, when the current empty container inventory exceeds value U, the empty container inventory should be repositioned outward so that the empty container inventory can be restored to the threshold range. Eq. (26) indicates that, when the empty container inventory is below value D, the empty container inventory should be replenished by repositioning inward or renting containers.

Similarly, the following constraints are added to construct a joint optimization model for empty container repositioning and storage under the periodic inventory control strategy:

$$Q_i^t = \begin{cases} S_i^t - ST_i^t = \sum_{j \in P} \beta_i^t QP_{ij}^t + \sum_{m \in H} \alpha_m^t QE_{im}^t + \beta_i^t QL_i^t, S_i^t \ge ST_i^t \\ 0, S_i^t < ST_i^t \end{cases}, \forall t \in T, \forall i \in P \tag{27}$$

$$\sum_{m\in H}\lambda_{im}\alpha_m^t QE_{im}^t + \sum_{j\in P}\beta_i^t QP_{ij}^t + \beta_i^t QL_i^t \le \max\left(S_i^t - ST_i^t, 0\right), \forall t\in T, \forall i\in P \tag{28}$$

$$\sum_{m\in H}\lambda_{im}\alpha_m^t QE_{im}^t + \sum_{j\in P}\beta_i^t QP_{ij}^t \le \max\left(ST_i^t - S_i^t, 0\right), \forall t\in T, \forall i\in P \tag{29}$$

Eqs. (27)–(29) indicate that, when the port empty container inventory does not reach the maximum value S, shipping company needs to reposition empty containers from ports and inland terminals and empty container leasing. If the empty container inventory exceeds the maximum value , the shipping company needs to reposition empty containers outward. The repositioning direction is ports within the cluster and inland terminals within port coverage.

4.5. Opportunity constraints in non-stationary demand

In an uncertain environment, the supply and demand of empty containers are random variables; thus, the above constraint expressions (12)–(15) and (18) related to the supply of and demand for empty containers need to be rewritten as chance constraint expressions. Set the lower bound of the probability of each of the above five opportunity constraints as ε. After transformation, the five opportunity constraint expressions are as shown in Eqs. (30)–(34):

$$\Pr\left\{\beta_i^t\left(\sum_{j\in P}QP_{ij}^t + \sum_{m\in H}\lambda_{im}QE_{im}^t\right) \le ST_i^{t-1} + \sum_{j\in P}Iq_{ji}^{t-1} - \sum_{m\in H}\lambda_{im}QH^{t-1} - \sum_{j\in P}Oq_{ij}^t\right\} \ge \varepsilon \quad \forall i\in P, \forall t\in T \tag{30}$$

$$\Pr\left\{-\sum_{j\in P}\beta_j^t QP_{ji}^{t-1} - \sum_{m\in H}\alpha_m^t\lambda_{im}QE_{mi}^{t-1} - \beta_i^t QL_i^t \le ST_i^{t-1} + \sum_{j\in P}Iq_{ji}^{t-1} - \sum_{m\in H}\lambda_{im}QH^{t-1} - \sum_{j\in P}Oq_{ij}^t\right)\right\} \ge \varepsilon \quad \forall i\in P, \forall t\in T \tag{31}$$

$$\Pr\left\{\alpha_m^t\left(\sum_{i\in P}\lambda_{im}QE_{mi}^t + \sum_{n\in H}\gamma_{mn}QX_{mn}^t\right) \le ST_m^{t-1} + \sum_{i\in P}\lambda_{im}QH_{im}^{t-1} - \sum_{i\in P}DH_{mi}^t\right\} \ge \varepsilon \quad \forall i\in P, \forall t\in T \tag{32}$$

$$\Pr\left\{-\sum_{i\in P}\beta_i^t\lambda_{im}QE_{im}^{t-1} - \sum_{n\in H}\beta_n^t\gamma_{mn}QX_{nm}^t - \alpha_m^t QL_m^t \le ST_m^{t-1} + \sum_{i\in P}\lambda_{im}QH_{im}^{t-1} - \sum_{i\in P}DH_{mi}^t\right)\right\} \ge \varepsilon \quad \forall i\in P, \forall t\in T \tag{33}$$

$$\Pr\left\{QP_{ij}^t - F_{ij}^t \le -Oq_{ij}^t - \sum_{m\in H}\lambda_{im}DH_{mi}^t\right\} \ge \varepsilon, \forall i\in P, \forall t\in T \tag{34}$$

The chance constraint expressions (30) and (31) indicate that, in decision period t, the number of empty containers repositioned from the residual container node (including ports and inland container terminals) to the shortage node must be less than the number of empty containers available in the residual container node at this point in time with a probability of not less than ε. The chance constraint expressions (32) and (33) indicate that, in decision period$t$, the number of empty containers repositioned from the residual node to the shortage node and number of renting containers repositioned from the shortage node (including ports and inland container terminals) must be less than the number of empty containers available at the shortage node at this point in time with a probability of not less thanε. Eq. (34) expresses the sum of the volume of empty containers repositioned between ports, volume of full containers transported, and volume of full containers transported to the port from the terminal, which must be less than the total capacity of the ship with a probability of not less thanε.

5. Deterministic transformation of stochastic joint optimization model based on distributed robust optimization

The supply and demand of empty containers are difficult to predict accurately; thus, this study adopts the distributed robust chance constraints to handle the above chance constraints. The following notations are added:

ζ_m^t : Net inflow of empty containers of hinterland terminal$m\in H$ in period $t\in T$, which can be expressed as:

$$\zeta_m^t = ST_m^{t-1} + \sum_{i\in P}\lambda_{im}QH_{im}^{t-1} - \sum_{i\in P}\lambda_{im}DH_{im}^t, \forall m\in H, \forall t\in T \tag{35}$$

ζ_i^t : Net inflow of empty containers of port $i\in P$ in period$t\in T$, which can be expressed as:

$$\zeta_i^t = ST_i^{t-1} + \sum_{j\in P}Iq_{ji}^{t-1} - \sum_{m\in H}\lambda_{im}QH_{im}^{t-1} - \sum_{j\in P}Oq_{ij}^t, \forall i\in P, \forall t\in T \tag{36}$$

$\zeta_i'^t$: Sum of the number of full containers shipped out of port $i\in P$in period$t\in T$, which can be expressed as:

$$\zeta_i'^t = Oq_{ij}^t + \sum_{m\in H}\lambda_{im}DH_{mi}^t \tag{37}$$

μ_m^t : Expectation of net inflow $\zeta_m^t$of empty containers of hinterland terminal $m\in H$in period$t\in T$;

σ_{tm}^2 : Variance of net inflow $\zeta_m^t$of empty containers of hinterland terminal $m\in H$in period$t\in T$;

μ_i^t : Expectation of net inflow $\zeta_m^t$of empty containers of port $i\in P$in period$t\in T$;

σ_{ti}^2 : Variance of net inflow $\zeta_m^t$of empty containers of port $i\in P$in period$t\in T$;

$\mu_i'^t$: Expectation of the sum $\zeta_i'^t$of the number of full containers shipped out of port $i\in P$in period$t\in T$;

$\sigma_{ti}'^2$: Variance of the sum $\zeta_i'^t$of the number of full containers shipped out of port $i\in P$in period$t\in T$;

ψ : Set of probability distributionPP.

Rewrite the chance constraints as distributed robust optimization chance constraints as follows:

$$\underset{PP\in\psi}{minPP}\left\{\beta_i^t\left(\sum_{j\in P}QP_{ij}^t+\sum_{m\in H}\lambda_{im}QE_{im}^t\right)\le\zeta_i^t\right\}\ge\varepsilon,\forall i\in P,\forall t\in T \tag{38}$$

$$\underset{PP\in\psi}{\min}PP\left\{-\sum_{j\in P}\beta_j^tQP_{ji}^{t-1}-\sum_{m\in H}\alpha_m^t\lambda_{im}QE_{mi}^{t-1}-\beta_i^tQL_i^t\le\zeta_i^t\right)\right\}\ge\varepsilon,\forall i\in P,\forall t\in T \tag{39}$$

$$\underset{PP\in\psi}{minPP}\left\{\alpha_m^t\left(\sum_{i\in P}\lambda_{im}QE_{mi}^t+\sum_{n\in H}\gamma_{mn}QX_{mn}^t\right)\le\zeta_m^t\right\}\ge\varepsilon,\forall i\in P,\forall t\in T \tag{40}$$

$$\underset{PP\in\psi}{\min}PP\left\{-\sum_{i\in P}\beta_i^t\lambda_{im}QE_{im}^{t-1}-\sum_{n\in H}\beta_n^t\gamma_{mn}QX_{nm}^t-\alpha_m^tQL1_m^t\le\zeta_m^t\right)\right\}\ge\varepsilon,\forall i\in P,\forall t\in T \tag{41}$$

$$\underset{PP\in\psi}{minPP}\left\{QP_{ij}^t-F_{ij}^t\le-\zeta_i'^t\right\}\ge\varepsilon,\forall i\in P,\forall t\in T \tag{42}$$

Eqs. (38)– (42) represent the chance constraints in the worst-case distribution of $\zeta_m^t,\zeta_i^t,\zeta_i'^t$. Constraint (38) is used next as an example of a deterministic transformation.

Let$S=\beta_i^t\left(\sum_{j\in P}QP_{ij}^t+\sum_{m\in H}\lambda_{im}QE_{im}^t\right)-\zeta_i^t$; then, there is

$$E(S)=\beta_i^t\left(\sum_{j\in P}QP_{ij}^t+\sum_{m\in H}\lambda_{im}QE_{im}^t\right)-\mu_i^t \tag{43}$$

Thus, the following relationship exists:

$$\frac{S-E(S)}{\sqrt{\sigma_{ti}^2}}\le\frac{0-E(S)}{\sqrt{\sigma_{ti}^2}}\Rightarrow\frac{S-E(S)}{\sqrt{\sigma_{ti}^2}}\le\frac{\mu_i^t-\beta_i^t\left(\sum_{j\in P}QP_{ij}^t+\sum_{m\in H}\lambda_{im}QE_{im}^t\right)}{\sqrt{\sigma_{ti}^2}} \tag{44}$$

$$\Pr\left(\frac{S-E(S)}{\sqrt{\sigma_{ti}^2}}\le\frac{\mu_i^t-\beta_i^t\left(\sum_{j\in P}QP_{ij}^t+\sum_{m\in H}\lambda_{im}QE_{im}^t\right)}{\sqrt{\sigma_{ti}^2}}\right)\ge\varepsilon \tag{45}$$

$$K^{-1}(\varepsilon)\le\frac{\mu_i^t-\beta_i^t\left(\sum_{j\in P}QP_{ij}^t+\sum_{m\in H}\lambda_{im}QE_{im}^t\right)}{\sqrt{\sigma_{ti}^2}} \tag{46}$$

$$K^{-1}(\varepsilon)\sqrt{\sigma_{ti}^2}\le\mu_i^t-\beta_i^t\left(\sum_{j\in P}QP_{ij}^t+\sum_{m\in H}\lambda_{im}QE_{im}^t\right) \tag{47}$$

$$\beta_i^t\left(\sum_{j\in P}QP_{ij}^t+\sum_{m\in H}\lambda_{im}QE_{im}^t\right)\le\mu_i^t-K^{-1}(\varepsilon)\sqrt{\sigma_{ti}^2} \tag{48}$$

Based on the derivation of Ghaoui et al. [27],$K(\varepsilon)=\sqrt{\frac{1-\varepsilon}{\varepsilon}}$, the above inequality can be transformed into

$$\beta_i^t\left(\sum_{j\in P}QP_{ij}^t+\sum_{m\in H}\lambda_{im}QE_{im}^t\right)\le\mu_i^t-\sqrt{\frac{\varepsilon}{1-\varepsilon}}\sqrt{\sigma_{ti}^2} \tag{49}$$

Then, the chance constraint equations can all be transformed into deterministic constraints in the manner described above, as shown in Eqs. (50)–(54).

$$\beta_i^t\left(\sum_{j\in P}QP_{ij}^t+\sum_{m\in H}\lambda_{im}QE_{im}^t\right)\le\mu_i^t-\sqrt{\frac{\varepsilon}{1-\varepsilon}}\sqrt{\sigma_{ti}^2},\forall i\in P,\forall t\in T \tag{50}$$

$$\sqrt{\frac{\varepsilon}{1-\varepsilon}}\sqrt{\sigma_{ti}^2}-\mu_i^t\le\sum_{j\in P}\beta_j^tQP_{ji}^{t-1}+\sum_{m\in H}\alpha_m^t\lambda_{im}QE_{mi}^{t-1}+\beta_i^tQL_i^t,\forall i\in P,\forall t\in T \tag{51}$$

$$\alpha_m^t\left(\sum_{i\in P}\lambda_{im}QE_{mi}^t+\sum_{n\in H}\gamma_{mn}QX_{mn}^t\right)\le\mu_m^t-\sqrt{\frac{\varepsilon}{1-\varepsilon}}\sqrt{\sigma_{tm}^2},\forall m\in H,\forall t\in T \tag{52}$$

$$\sqrt{\frac{\varepsilon}{1-\varepsilon}}\sqrt{\sigma_{tm}^2}-\mu_m^t\le\sum_{i\in P}\beta_i^t\lambda_{im}QE_{im}^{t-1}+\sum_{n\in H}\beta_n^t\gamma_{mn}QX_{nm}^t+\alpha_m^tQL_m^t,\forall m\in H,\forall t\in T \tag{53}$$

$$\sqrt{\frac{\varepsilon}{1-\varepsilon}}\sqrt{\sigma_{ti}'^2}+\mu_i'^t\le F_{ij}^t-QP_{ij}^t,\forall i,j\in P,\forall t\in T \tag{54}$$

6. Hybrid algorithm design based on dynamic programming and simulated annealing algorithms

6.1. Algorithm design ideas

Due to the stage, dynamic, and stochastic characteristics of the empty container repositioning problem reflected in the joint optimization model constructed in this paper, this model is a nonlinear mixed integer programming model and is established in the case of non-stationary demand, making the problem more complex to solve. Its complexity is mainly reflected in the following aspects. The first is the length of the decision-making cycle. The research cycle of this paper is 24 cycles, which is longer than that in the existing research. The problems that need to be solved in each cycle can be regarded as a separate module. The decision of the former module will directly affect the optimization decision of the latter module. If a certain item in the repositioning scheme changes, then the overall decision-making scheme will be qualitatively changed later. In addition, the iteration during the multi week period will also directly affect the calculation time. The second reason is that there are many decisions needed, including the repositioning volume between ports, between ports and terminals within the coverage area, the container rental volume of ports and terminals. At the same time, the derivative variable part also needs to consider the remaining inventory of empty containers in each cycle, which are all problems that need to be decided in solving. Thirdly, due to the randomness of the empty container repositioning process, it is necessary to reflect the handling of randomness and uncertainty in the calculation process. Finally, due to the introduction of a threshold for empty container inventory, it is necessary to ensure that the inventory of empty containers or the inventory of empty containers after container repositioning or renting operations remains within the threshold range. The problem adds threshold constraints on the basis of the original decision, making the optimization of decision variables more complex.

Usually, dynamic programming is used for the study of empty container repositioning. However, due to the complexity of the problem in this paper, it is impossible to obtain the optimal solution within a reasonable time. Traditional heuristic algorithms, hybrid heuristic algorithms, or CPLEX exact solution methods cannot better solve the model. Therefore, this paper designs a hybrid algorithm combining dynamic programming and simulated annealing based on the characteristics of the research problem. In hybrid algorithms, dynamic programming algorithm is used as the main algorithm, and nested simulated annealing algorithm is used. The main algorithm uses the inventory of empty containers in each cycle as a dynamic transfer equation, and uses dynamic programming to solve the stage and dynamics of the problem. The simulated annealing algorithm is used to solve the optimal decision for each stage. At the same time, in the process of generating new solutions at each stage, in order to avoid the

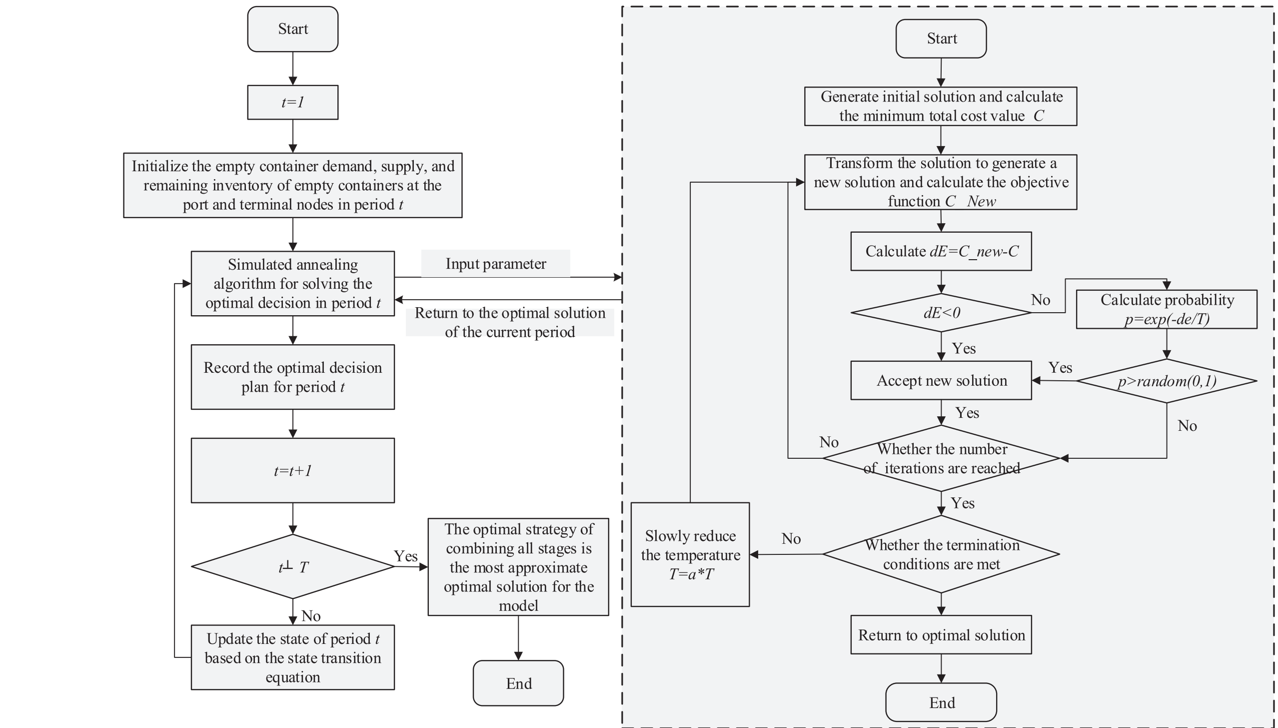

Fig. 3. Flowchart of the hybrid algorithm.

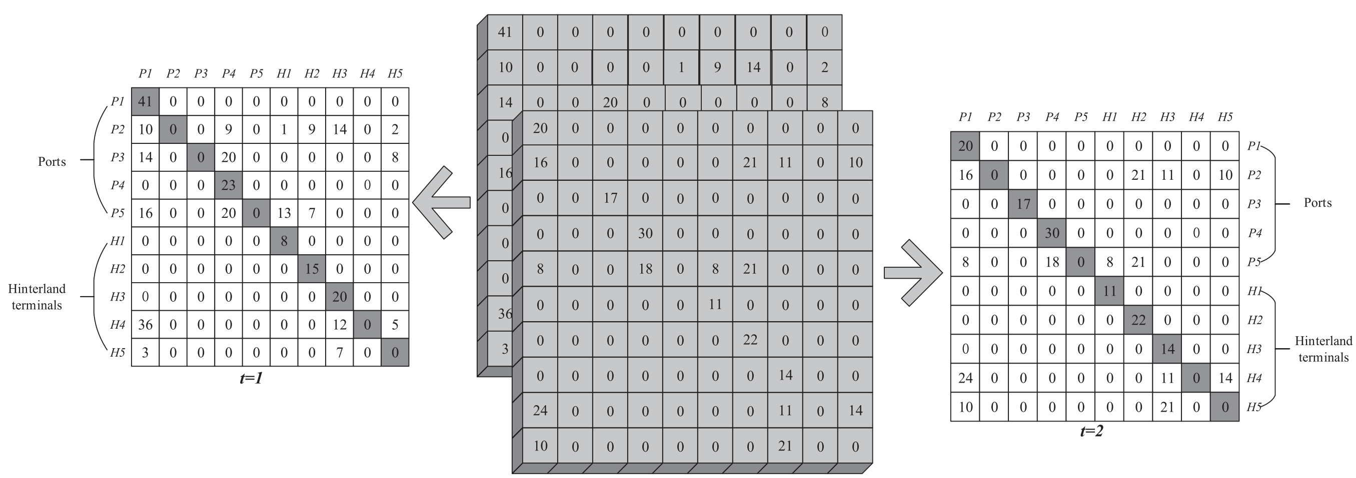

Fig. 4. Coding diagram.

reverse flow of empty container resources and the generation of a large number of infeasible solutions, that is, the flow from surplus container points to shortage points, we innovatively designed new heuristic rules to improve the overall algorithm's solving efficiency. The algorithm flowchart is shown in Fig. 3.

Let $SS_t = \left\{ST_m^t, ST_i^t, QP_{ij}^t, QE_{im}^t, QL_m^t, QL_i^t, QX_{mn}^t \forall t \in T, i,j \in P, m,n \in H\right\}$, then, the dynamic transfer equations are shown below.

$$SS_t(ST_i^t) = SS_t(ST_i^{t-1}) + \beta_i^t QL_i^t + \omega_i^t + \sum_{j\in P}\beta_i^t QP_{ji}^{t-1} + \sum_{m\in H}\alpha_m^t \lambda_{im} QE_{im}^t - \beta_i^t\left(\sum_{j\in P} QP_{ij}^t + \sum_{m\in H}\lambda_{im} QE_{im}^t\right), \forall t \in T, \forall i \in P \tag{56}$$

$$SS_t(ST_m^t) = SS_t(ST_m^{t-1}) + \sum_{i\in P}\lambda_{im} QH_{im}^{t-1} - \sum_{j\in P} DH_{mi}^t + \sum_{i\in P}\beta_i^t \lambda_{im} QE_{im}^t + \sum_{n\in H}\alpha_n^t \gamma_{mn} QX_{nm}^t + \alpha_m^t\left(QL_m^t - \sum_{n\in H}\gamma_{mn} QX_{mn}^t - \sum_{i\in P}\lambda_{im} QH_{im}^{t-1}\right), \forall t \in T, \forall m \in H \tag{55}$$

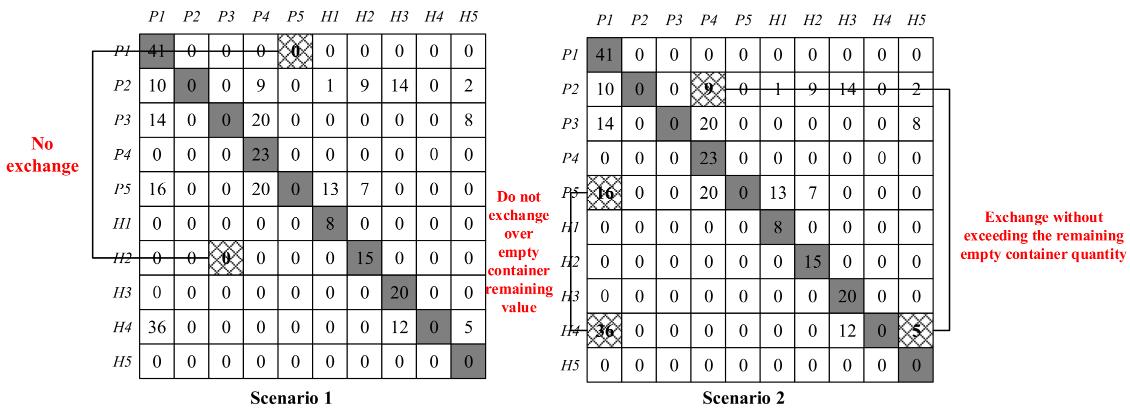

Fig. 5. Schematic diagram of solution transformation under Cases 1 and 2.

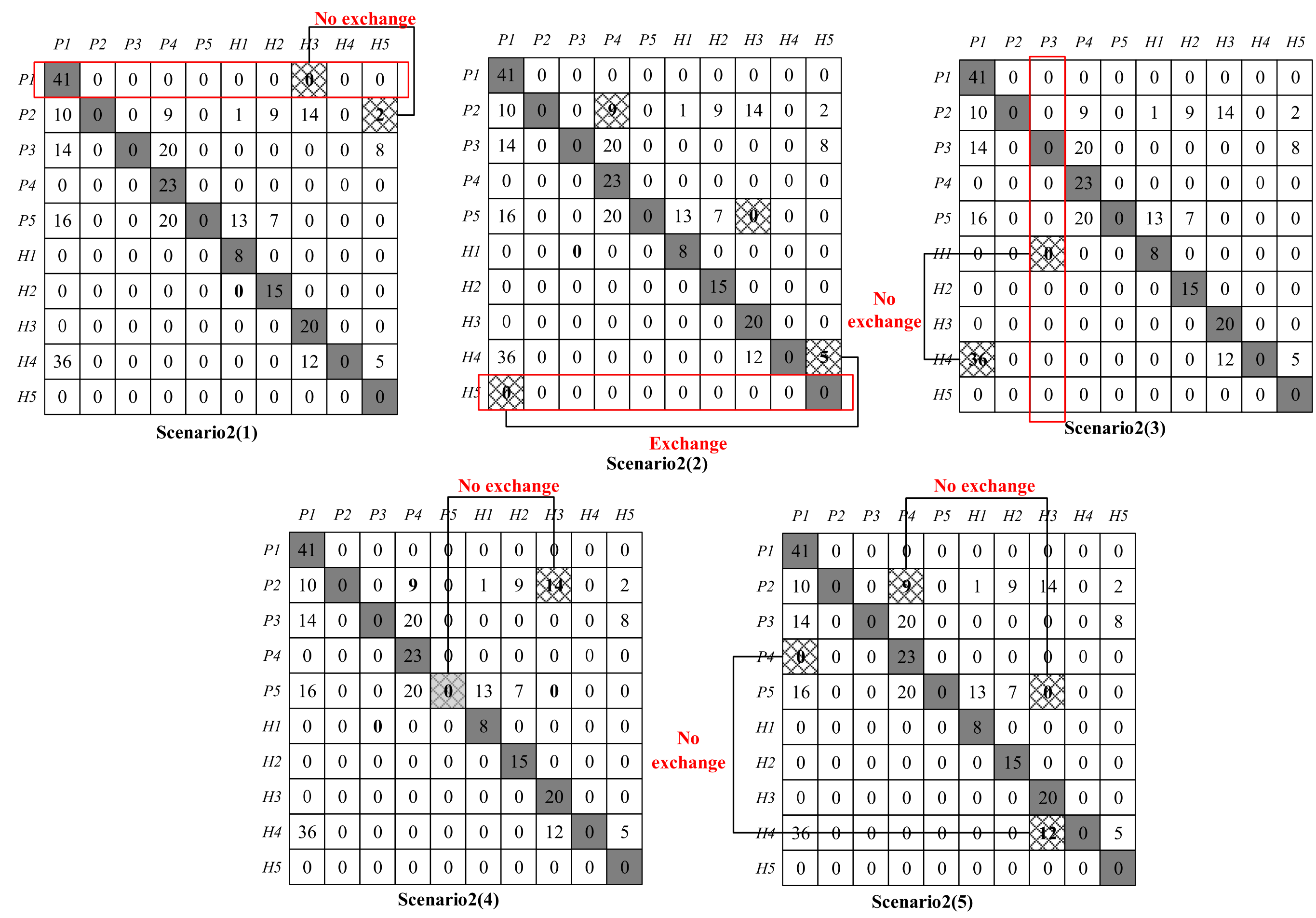

Fig. 6. Schematic diagram of solution transformation under Case 3.

6.2. Coding design

In this study, the natural number coding method is used. The values in the matrix represent the volume of empty container repositioning and volume of renting containers. The value of the volume of empty container repositioning is 0 if it is not in the coverage area or inaccessible between the ports and terminals, and the diagonal position represents the volume of renting containers in the port and terminals. Fig. 4 shows the representation of the coding design.

6.3. New solution generation method

A crossover operation is performed on the current solution to generate new solutions (i.e., a new empty container repositioning scheme and a renting scheme). First, the crossover strategy is used, which applies the method of generating random numbers to crossover two elements in the matrix to generate a new empty container repositioning solution. Two random positions are generated in the matrix. The numbers corresponding to the randomly generated positions then have three possible scenarios (i.e., the values in both positions are 0, or 0 and a non-zero integer, or a non-zero integer and a non-zero integer). As shown in Figs. 5 and 6, the crossover operation for each of these scenarios cam be performed three ways:

Scenario 1: the values of the two random positions are both 0, indicating that they may not be in the coverage or not renting containers, or the two ports are both sufficient ports; they do not need to accept the repositioning of containers and do not carry out the crossover at this time;

Scenario 2: two random positions on the value of the non-zero integer and non-zero integer, to determine whether the exchange value exceeds the value of the remaining empty containers $W(t,i)$. If not exceeded, the values at the two random positions are exchanged; otherwise, they are not;

Scenario 3: the values in the two random positions are 0 and non-zero integers; at this time, the following cases may occur:

(1) The values in the row where the value of random position 0 is located are all 0 except for the diagonal value, which indicates that the node in this row is a demand node, and it cannot be repositioned outward, so it is not exchanged;

(2) Random position 0 value in the line where all the values are 0, indicating that the node in this line is a node of the remaining containers and can be repositioned to the outside; at this time, the exchange is performed;

(3) The value of a column where the value of random position 0 is located is all 0, which indicates that this column node is a supply node and needs to accept incoming containers from other nodes; at this time, no exchange is performed;

(4) If random position 0 is located on the diagonal of the matrix, then the corresponding node is a supply node and does not need to rent a container; at this time, no exchange is made;

(5) The existence of other values in the column where the value of random position 0 is located is also not exchanged, because the value of 0 indicates that it is not in the coverage or the demand port, or the supply port does not need to accept the repositioning from other nodes; in this case, no exchange is performed.

The above solution transformation operation avoids repositioning more than the inventory at each node when generating a new solution, as well as avoids the flow of empty containers from the shortage node to the residual node.

6.4. Algorithm parameter selection

The hybrid algorithm includes simulated annealing and dynamic programming. Firstly, as the main algorithm, the initial lower bound of the probability for chance constraint is set to 0.5, the initial demand mean is randomly generated within the interval of [150,500] in non-stationary situations. The variance of demand within each cycle is generated within the range of [20,50]. The termination condition of overall iteration depends on the number of decision cycles.

In the annealing process, we set the initial temperature T_0 value to 1000 and the algorithm termination temperature to 0.01. The overall annealing strategy of the algorithm adopts a temperature drop function, $T_{\theta+1} = \alpha T_\theta$, which θ represents the current state and the annealing rate α, taken as 0.9. Set the number of internal cycles at each temperature as$1000 - \frac{T}{2}$. The hybrid simulated annealing algorithm generates new energy values for each solution transformation, and the difference between the new solution and the original solution is set toΔE. If$\Delta E < 0$, the algorithm accepts a new solution with a probability of 1; if$\Delta E > 0$,

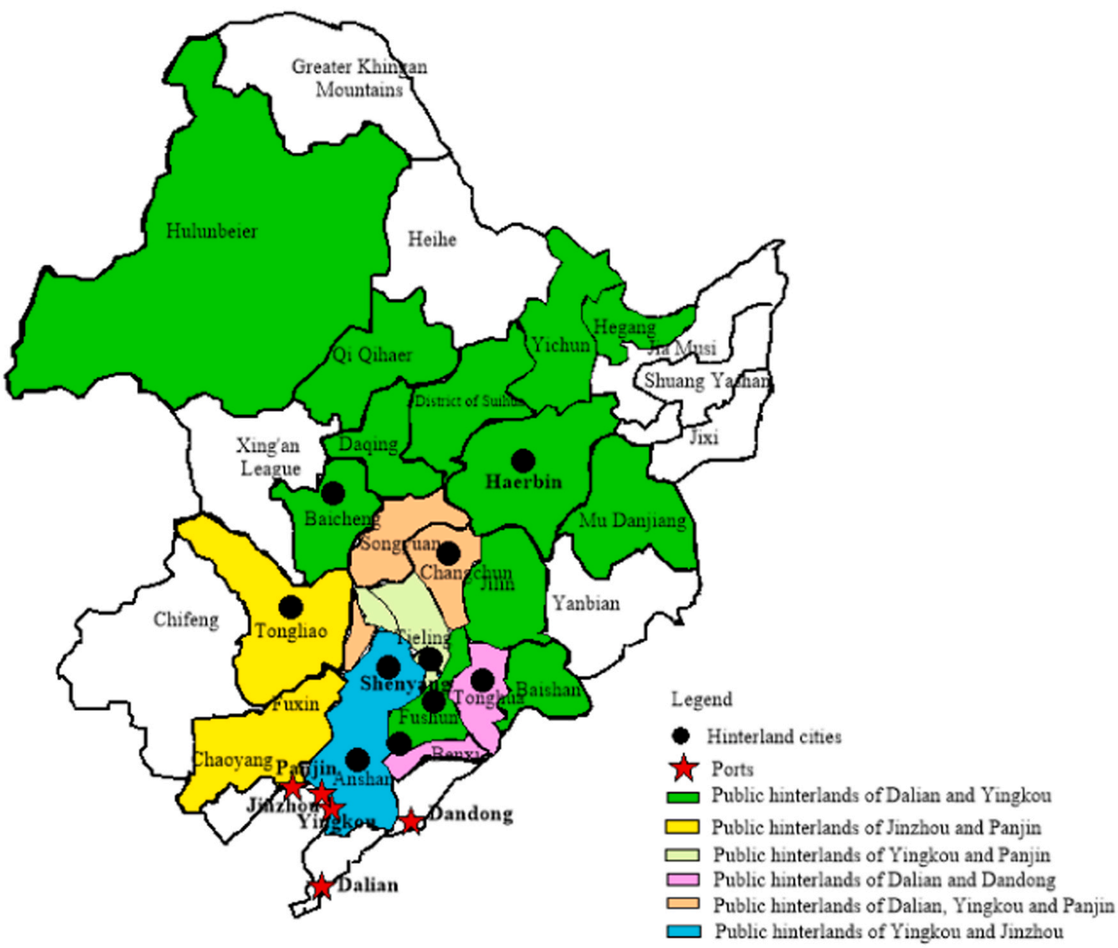

Fig. 7. Schematic diagram of selected ports and hinterland cities.

Table 4
Unit empty container repositioning cost between nodes.

Cost (yuan)	Dalian	Yingkou	Dandong	Panjin	Jinzhou	Shenyang	Anshan	Haerbin	Changchun	Tonghua
Dalian	0	97.05	126	115.15	188.8	682.56	537.84	1828.44	1216.98	1121.4
Yingkou	97.05	0	136.16	23.7	67.48	811.8	459.45	1345.86	861.48	790.92
Dandong	126	136.16	0	110.16	151	437.76	424.8	-	-	495
Panjin	92.16	23.7	110.16	0	56.68	865.8	501.3	-	-	-
Jinzhou	188.8	67.48	151	56.68	0	422.46	385.2	-	-	-
Shenyang	682.56	811.8	437.76	865.8	422.46	0	464.85	-	-	-
Anshan	537.84	459.45	424.8	501.3	385.2	464.85	0	-	-	-
Haerbin	1828.44	1345.86	-	-	-	-	-	0	489.96	970.38
Changchun	1216.98	861.48	-	-	-	-	-	489.96	0	477.54
Tonghua	1121.4	790.92	495	-	-	-	-	970.38	477.54	0

the algorithm accepts the new solution with probability$e^{\frac{-\Delta E}{T}}$.

The overall algorithm starts with the initial parameters input from the dynamic programming algorithm, calculates the optimal decision for each decision cycle using the simulated annealing algorithm, and returns the optimal decision value to the current decision stage in the dynamic programming algorithm. Then, the next iteration is performed through the dynamic transition equation until all decision cycles are calculated.

7. Numerical experiment

7.1. Numerical examples selection

This study uses the Liaoning coastal port cluster – Northeast hinterland as an example to examine the optimization of empty container repositioning within the port cluster considering inventory control strategies. The Liaoning coastal port cluster includes five ports, Dalian, Dandong, Yingkou, Panjin, and Jinzhou Ports [28], and container terminals in five hinterland cities, Shenyang, Anshan, Changchun, Tonghua, and Harbin (Fig. 7). Unit empty container repositioning cost between nodes of ports and terminals is shown in Table 4.

The coverage range between ports and hinterland terminals, as well as the accessibility between stations, can be represented by the matrix of λ_{im} and γ_{mn} as

$$\lambda_{im} = \begin{pmatrix} 1,1,1,1,1 \\ 1,1,0,0,1 \\ 1,1,0,0,1 \\ 1,1,0,0,0 \\ 1,1,0,0,0 \end{pmatrix}, \gamma_{mn} = \begin{pmatrix} 0,1,0,0,0 \\ 1,0,0,0,0 \\ 0,0,0,1,1 \\ 0,0,1,0,1 \\ 0,0,1,1,0 \end{pmatrix}$$

The data in this paper is sourced from Dalian Port and existing literature references. The decision-making cycle is 24 weeks, with each cycle lasting 7 days. Owing to the non-stationary nature of the empty container demand set in this study, the demand is replaced by the opportunity constraint and introduction of mean and variance. The costs under stable demand are also calculated. In this case, the demand of each port follows the normal distribution, and Dalian Port~$(300,20^2)$, Yingkou Port~$(200,20^2)$, Dandong Port~$(100,20^2)$, Jinzhou Port~$(100,20^2)$, Panjin Port~$(80,20^2)$ are set.

7.2. Calculation results

Table 5 shows the calculation results for various scenarios. For convenience, quantitative inventory control strategies are represented by (D, U), periodic inventory control strategies are represented by (T, S), and models without inventory control strategies are represented by Basic. The repositioning cost is recorded as C1, renting cost as C2, storage cost as C3, and total cost as C.

The calculation results show that, regardless of whether the empty container threshold is set, the cost values under non-stationary empty container demand are higher than those under stable demand. Uncertain changes will also lead to an increase in renting volume for shipping companies, which in turn leads to an increase in total costs. In contrast, under stable demand, the changes in various cost values are relatively small. If the accurate demand for empty containers can be understood, then the shipping company can also more accurately understand the repositioning, renting volume for each port in all periods. Furthermore, the table shows that whether the demand is non-stationary or stationary, the empty container inventory threshold has a better cost control effect under non-stationary demand, which aligns with shipping companies' actual operation situation. In practical situations, shipping companies often face random empty container demand, making it difficult to accurately grasp the empty container demand in each period. Setting an empty container inventory threshold can effectively address the randomness of empty container demand, thereby more effectively reducing total costs.

From the calculation results in the table, it can be seen that under non-stationary demand, the periodic inventory control strategy can reduce the total cost of the shipping company by 6.44 %, while the quantitative inventory control strategy is only 4.90 %. Under stationary demand, quantitative inventory control strategy can reduce the total cost of shipping companies by 9.05 %, which is more outstanding compared to the 7.03 % performance of periodic inventory control strategy. If a shipping company has sufficient confidence in the demand for empty containers during the decision-making period, it can adopt a quantitative inventory control strategy to reduce costs and increase efficiency. In situations where the demand for empty containers is difficult to predict and uncertain, a periodic inventory control strategy can be chosen to reduce the total cost.

This conclusion is related to the characteristics of the two inventory control strategies and non-stationary demand. Under non-stationary

Table 5
Calculation results under different scenarios.

Cost (ten thousand yuan)	Under non-stationary demand			Under stationary demand		
	(D, U)	(T, S)	Basic	(D, U)	(T, S)	Basic
C1	201.2	187.46	213.5	151.2	149.51	168.3
C2	585.83	543.56	621.77	510.56	489.06	579.34
C3	156.4	197.2	156.8	110.33	150.67	101.29
C	943.43	928.22	992.07	772.09	789.24	848.93
Gap of C	4.90 %	6.44 %	-	9.05 %	7.03 %	-

Gap of C={Basic-(D,U)(or (T,S))}/Basic

Table 6
Cost values under different lower bounds.

Cost (ten thousand yuan)		Under non-stationary demand (D, U)	(T, S)	Basic	Under stationary demand (D, U)	(T, S)	Basic
0.9	C1	234.6	205.7	251.1	189.09	171.43	194.21
	C2	616.7	587.88	657.16	541.6	512.57	613.4
	C3	189.05	223.2	193.2	131.8	170.9	128.09
	C	1040.35	1016.78	1101.46	862.49	854.9	935.7
0.7	C1	212.55	192.1	232.14	163.78	158.7	177.53
	C2	592.3	556.42	634.65	523.03	500.8	591.26
	C3	179.2	211.66	170.05	125.9	162.1	114.4
	C	984.05	960.18	1036.84	812.71	821.6	883.19
0.3	C1	185.1	165.4	200.01	136.4	132.5	149.1
	C2	553.19	521.3	594.85	471.32	460.57	542.2
	C3	127.8	179.01	131.9	102.8	125.91	89.78
	C	866.09	865.71	926.76	710.52	718.98	781.08

demand, shipping companies find it difficult to grasp the specific demand for empty containers, and the emergence of uncertain demand is usually unstable and urgent. If shipping companies cannot provide empty container resources, they need to rent or reposition containers. If the demand cannot be met, customers will be lost and revenues will be lost. In this situation, the periodic inventory control strategy, due to its own characteristics, only needs to replenish the empty containers according to a fixed cycle. This means that by calculating the threshold for empty container inventory, shipping companies can maintain a sufficient amount of empty container inventory to meet sudden container demand during each decision-making period, which can reduce the rental volume and thus reduce the total cost of empty container management. In addition, the periodic inventory control strategy involves conducting inventory checks on a fixed cycle, resulting in reduced management costs such as inventory checks. This is suitable for shipping companies to respond to sudden empty container demands.

Under the stationary demand, the shipping company has sufficient confidence in the demand for empty containers at a certain stage, and only needs to replenish empty containers according to the stationary demand of each cycle. At this point, shipping companies do not need to keep a large number of empty containers at the port, as they have a clear understanding of the demand for empty containers. Therefore, adopting a quantitative inventory control strategy and managing empty container inventory according to the upper and lower limits of the empty container inventory threshold will not result in a large amount of empty container resources piling up at the port, leading to resource waste and excess inventory cost expenditures.

7.3. Sensitivity analysis

(1) Changes in lower bound of the probability for chance constraint and variance

To explore the impact of changes in uncertain parameters, two sensitivity experiments are conducted. Different lower bound parameters and variances are selected and substituted into the model for calculation (Table 6 and 7).

The results in Table 6&7 show that, as lower bounds or variances continue to increase, cost values under both non-stationary and stationary demand constantly increase. This is mainly determined by the equivalent constraints after transformation. As they increase, the

Table 7
Calculation results of each cost value under variance change.

Cost (ten thousand yuan)		Under non-stationary demand (D, U)	(T, S)	Basic	Under stationary demand (D, U)	(T, S)	Basic
20 %	C1	211.42	191.1	245.7	168.48	161.67	177.12
	C2	597.05	560.3	650.9	523.2	499.08	602.5
	C3	164.87	205.6	176.92	124.1	178.63	120.04
	C	973.34	957	1073.52	815.78	839.38	899.66
40 %	C1	227.56	212.2	271.39	175.6	172.3	196.35
	C2	613.98	578.05	681.2	535.9	509.85	630.44
	C3	178.93	216.62	196.33	138.19	190.37	139.68
	C	1020.47	1007.32	1148.92	849.69	872.52	966.47
60 %	C1	241.02	227.33	299.4	190.2	188.6	230.01
	C2	629.3	590.8	703.3	546.6	518.65	649.57
	C3	179.9	231.4	221.71	146.97	210.3	139.45
	C	1050.22	1049.53	1224.41	883.77	917.55	1019.03
80 %	C1	254.55	239.09	310.76	201.1	187.5	246.7
	C2	637.71	603.6	719.9	557.98	531.6	656.32
	C3	198.46	245.38	242.5	159.75	222.54	156.9
	C	1090.72	1088.07	1273.16	918.83	941.64	1059.92
100 %	C1	268.19	248.8	323.91	219.09	192.3	261.5
	C2	658.2	613.97	725.6	569.22	544.79	672.68
	C3	210.03	258.56	266.74	163.8	239.4	172.16
	C	1136.42	1121.33	1316.25	952.11	976.49	1106.34

Table 8
Cost values under the change of accessibility between terminals.

Cost (ten thousand yuan)		Under non-stationary demand (D, U)	(T, S)	Basic	Under stationary demand (D, U)	(T, S)	Basic
γ_{mn}	C1	201.2	187.46	213.5	151.2	149.51	168.3
	C2	585.83	543.56	621.77	510.56	489.06	579.34
	C3	156.4	197.2	156.8	110.33	150.67	101.29
	C	943.43	928.22	992.07	772.09	789.24	848.93
γ'_{mn}	C1	245.6	221.34	277.4	200.9	189.44	205.7
	C2	600.5	572.13	680.98	542.52	513.25	613.4
	C3	179.07	229.8	188.63	147.61	195.5	169.88
	C	1025.17	1023.27	1147.01	891.03	898.19	988.98
GAP of total costs		8.66 %	10.24 %	15.62 %	15.40 %	13.8 %	16.50 %

decision variables used to meet demand after the opportunity constraint transformation also increase, thereby leading to an increase in the cost of the entire empty container repositioning system. Furthermore, regardless of the change in lower bound or variance, the optimization effect of the periodic inventory control strategy on total cost under non-stationary demand is better than that of the quantitative inventory control strategy, which is consistent with the abovementioned solution results.

(2) Changes in accessibility between inland terminals

The sea–land empty container repositioning network within the port cluster set up in this study includes a direct return mode between customers (i.e., empty container repositioning between inland terminals). Therefore, to explore the effectiveness of this mode, empty container repositioning between terminals is cancelled. The results are compared with those of the joint optimization model to verify the effectiveness of this mode. If no accessibility exists between terminals, the change in parameters is as follows. The calculation results of various cost values are shown in Table 8.

$$\gamma_{mn} = \begin{pmatrix} 0,1,0,0,0 \\ 1,0,0,0,0 \\ 0,0,0,1,1 \\ 0,0,1,0,1 \\ 0,0,1,1,0 \end{pmatrix} \Rightarrow \gamma'_{mn} = \begin{pmatrix} 0,0,0,0,0 \\ 0,0,0,0,0 \\ 0,0,0,0,0 \\ 0,0,0,0,0 \\ 0,0,0,0,0 \end{pmatrix}$$

The results in Table 8 show that, when the accessibility between inland terminals changes, shipping companies' total costs increases, regardless of whether the demand is stable. The total cost increase under non-stationary demand is higher than that under stationary demand, while the total cost under the threshold of empty container inventory is lower than that without the threshold. This indicates that when the accessibility between inland terminals is 0, the source of empty containers at inland terminals is only from the port, and the amount of empty container repositioning to the port also decreases. When the demand is non-stationary, the shipping company needs to meet the current demand for empty containers by renting a high number of containers. The threshold of empty container inventory can effectively reduce the fluctuation in total costs and reduce the impact of external environmental changes on shipping companies.

Comprehensively considering both inventory control strategies, each has its own advantages and disadvantages in different demand situations. Under non-stationary demand, if it is not reachable between terminals, a periodic inventory control strategy can significantly reduce the total cost of empty container management for shipping companies. This is also related to its larger empty container inventory threshold, which can greatly reduce the number of rented containers help control total cost. In the case of determined demand, if no accessibility exists between terminals, the quantitative inventory control strategy is more effective in controlling total costs. Therefore, for shipping companies, it is necessary to flexibly choose empty container inventory control strategies, reasonably establish empty container inventory thresholds, and make timely changes to strategies when the external environment changes to optimize the cost-control effect.

(3) Changes in demand distributions

To investigate the changes in total cost under different demand distributions, we select two other demand distributions, namely uniform distribution and Poisson distribution, and compared them with the normal distribution used in this paper. For the convenience of comparison, we select a uniform distribution of U~(0200), $\lambda = 200$ in Poisson distribution, and normal distribution N~ (200,20^2). The calculation results are shown in Table 9.

From the calculation results in the table, it can be seen that the total cost under normal distribution is the lowest regardless of whether the actual demand is non-stationary or stable, while the total cost value under Poisson distribution is the highest. When the parameter values are roughly the same, the normal distribution shows the most stable performance, while the Poisson distribution shows the most significant changes. This result is also related to the characteristics of each distribution itself, and indirectly indicates that if the demand fluctuates too much, it will lead to an expansion of the total cost. For shipping companies, maintaining a stable supply and demand of empty containers, or having a clear understanding of the demand for empty containers, plays an important role in reducing total costs.

Table 9
Total costs under the change of demand distributions.

Total cost (ten thousand yuan)	Under non-stationary demand			Under stationary demand		
	(D, U)	(T, S)	Basic	(D, U)	(T, S)	Basic
N	960.3	947.15	1089.2	805.1	821.4	916.6
U	977.5	960.8	1105.4	837.5	878.06	942.9
P	1024.7	981.9	1219.22	866.4	890.5	983.1

8. Conclusions

8.1. Concluding remarks

(1) In the case of non-stationary demand, joint optimization of empty container repositioning and inventory control can effectively reduce shipping companies' total container management costs. The quantitative and periodic inventory control strategies can reduce total costs by 4.9 % and 6.43 %, respectively. However, regardless of the inventory control strategy, joint optimization always has a significant effect.

(2) If shipping companies can accurately comprehend shippers' demand for empty containers within a certain period, then a quantitative inventory control strategy should be adopted for storing empty containers. If shipping companies are unable to roughly understand the demand for empty containers within a certain period of time, or face a large amount of temporary demand, then they should choose a periodic inventory control strategy to control empty containers address unpredictable demand and improve service levels. Essentially, the cost of renting containers is much higher than the cost of container repositioning and empty container inventory. In the case of non-stationary demand, the periodic inventory control strategy effectively suppresses and controls the demand for renting containers from shipping companies by maintaining a large amount of empty container inventory at ports, thereby achieving the goal of reducing total costs.

(3) Accessibility between inland terminals is extremely important for the supply of empty container resources for shipping companies. The findings in this study show that, if inland terminals are inaccessible, shipping companies will face a total cost increase of 8.66 % at least and 16.5 % at most, which undoubtedly increases shipping companies' costs for empty container management. In this situation, the periodic inventory strategy is better for reducing total costs by 10.24 % under non-stationary demand, while the quantitative strategy can reduce costs by 15.4 % under stable conditions. Therefore, shipping companies need to not only pay attention to the smoothness of maritime channels but also maintain accessibility between inland terminals, to ensure that empty containers from inland sources can be replenished to the port as much as possible.

8.2. Future work

(1) Empty container repositioning not only occurs within the regional port cluster, but also outside the port cluster. When the empty containers within the regional port group cannot meet the demand for empty containers, it is necessary to initiate empty container repositioning to other ports outside the port group. It is necessary to broaden the focus to the transportation of empty containers between different port groups, including the optimization of large-scale empty container transportation between domestic and international port groups. Due to the expansion of research scope, especially when it comes to the

repositioning of empty containers between international port clusters, multimodal transport needs to be considered as the main mode of repositioning of empty container resources. At the same time, the selection of transportation modes is carried out in different transportation sections, and the joint optimization of empty container repositioning and transportation mode selection is carried out to expand the joint optimization model constructed in this paper.

(2) This paper focuses on the TEU as the main research object, and in the future, foldable containers will be considered in the model. Folding containers, when not loaded with goods, occupy only a quarter of the volume of standard containers. With the same capacity, using foldable containers can transport a larger number of empty containers. When conditions permit, the use of foldable containers will greatly reduce the total cost of empty container management for shipping companies.

(3) In this paper, the equivalent transformation method is adopted to solve the impact of demand uncertainty. In the future, various methods can be used to address the issue of uncertainty, and comparative verification can be conducted. It will be further expanded to time-series analysis and stochastic optimization, mining the laws between empty container supply and demand, and providing a data foundation for joint optimization.

CRediT authorship contribution statement

Ying Huang: Validation, Software, Methodology. **Jiaxin Cai:** Writing – original draft, Validation, Software, Methodology, Conceptualization. **Zhihong Jin:** Writing – review & editing, Visualization. **Cuijie Diao:** Validation, Methodology, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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